9 **(a)** Draw the symbol for:

(i) a diode

[1]

(ii) a NOT gate.

[1]

(b) **(i)** Fig. 9.1 shows a digital circuit.

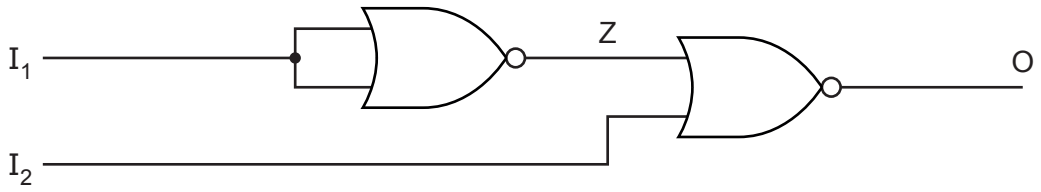


Fig. 9.1

Complete the truth table shown in Table 9.1.

Table 9.1

I_1	I_2	Z	O
0	0		
0	1		
1	0		
1	1		

[2]

(ii) State another single gate which is equivalent to the part of the circuit between I_1 and Z.

..... [1]

(c) Using **two** logic gates, design and draw a digital circuit with two inputs and two outputs which has the truth table shown in Table 9.2.

Use either the usual logic gate symbols or correctly labelled square boxes in your diagram.

Table 9.2

input 1	input 2	output 1	output 2
0	0	0	1
0	1	1	1
1	0	1	1
1	1	1	0

[4]

[Total: 9]

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